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Water Management Strategy in UAEs Unconventional Fields: A Case Study of Closed Loop Operations and Novel Treatment Technologies

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Abstract

As part of the UAE's commitment to Environmental Social Governance (ESG), unconventional exploration operations adhere to a strict zero-tolerance policy regarding spills. Consequently, closed-loop operations were instituted across all rigs and sites, ensuring no water is discharged into pits. This approach is supported by a thoroughly planned treatment and discharge strategy established before operations begin.

This paper will delineate the water management strategy, focusing on the closed-loop system and innovative treatment technologies that facilitate hydrogen sulfide (H_2S), removal without infusing chemicals into the produced fluid. By preventing both over- and under-treatment, this method not only enhances operational efficiency but also reduces costs. A major challenge encountered was managing and treating high temperature flowback water exceeding 200 degrees Fahrenheit.

Additionally, the paper will explore the technical and operational experiences in establishing the water management process for Zipper frac operations, where over 800,000 barrels of water were pumped. It includes the sourcing, storage, and treatment protocols, alongside modifications to testing and treatment equipment designed to handle high-temperature, high-pressure flowback water. The use of real-time, non-chemical hydrogen sulfide (H_2S), treatment methods reduced testing time by 50%.

Initiatives aiming for efficient and cost-effective water management solutions are being explored to facilitate the profitable development of unconventional gas and oil in the UAE. These initiatives strive to merge optimal performance and low-cost strategies with current regulations and environmental requirements to ensure sustainable operational security. The scope of the paper extends these closed-loop operations to metal extraction, applying circular economy principles to unconventional completion operations.

While the development of unconventional oil and gas resources has transformative impacts on the UAE and global energy markets, concerns about water availability in desert regions remain. The UAE, however, benefits from significant oil and gas reserves as well as abundant water resources—both surface and subterranean—and advanced local water treatment technologies and services. Moreover, the basin's

proximity to the sea and the stratigraphic setting of target unconventional reservoirs below known aquifers like Simsima and Um Radhuma provide notable advantages.

Introduction

One of the key challenges and opportunities in unconventional development is water management, which encompasses the entire water cycle from sourcing, transportation, storage, treatment, reuse, and disposal. Water management is critical for the environmental, social, and economic sustainability of unconventional operations, as it affects the operational efficiency, cost, safety, and environmental footprint of the projects. Water management is also influenced by the local context and regulations, such as the availability and quality of water sources, the demand and competition for water from other sectors, the environmental and social impacts of water use and disposal, and the legal and institutional frameworks governing water resources.

The UAE faces a unique set of challenges and opportunities in water management for unconventional development, given its arid climate, high population growth, and economic diversification. The UAE is committed to reducing its water consumption and increasing its water efficiency, as part of its national vision and strategy for water security and sustainability.

This paper sets out to detail the comprehensive water management strategy and practices implemented in UAE unconventional fields, emphasizing the integrated closed-loop operations, cutting-edge treatment technologies, and the application of circular economy principles. By examining each stage of water handling—from sourcing and storage to treatment and reuse—the discussion provides insights into how these approaches collectively minimize environmental impact and optimize operational efficiency.

A particular focus is placed on the deployment of novel treatment technologies that enable effective removal of challenging contaminants such as hydrogen sulfide (H_2S), without introducing additional chemicals into the produced fluids. These advancements not only improve the safety and reliability of water management processes but also contribute to cost savings and regulatory compliance.

Furthermore, the paper explores the challenges faced during high-pressure, high-temperature operations, drawing lessons from real-world applications and adjustments to equipment and protocols. It reviews the strategies for managing water throughout the lifecycle of unconventional wells, and how collaboration between technical teams facilitated the development of tailored solutions suitable for the UAE's unique geological and regulatory environment.

By sharing these experiences, the paper aims to highlight future opportunities for further enhancing water management, supporting sustainable development, and reinforcing the UAE's position as a leader in responsible and innovative unconventional resource extraction.

Water Management Strategy

The water management Strategy for the unconventional wells was designed to manage the large volumes of water injected into the wells, as illustrated in **Fig. 1** for water used in the zipper frac. This plan also addresses handling the substantial flowback water from the producer wells. Instead of disposing of this water in pits, which might cause environmental contamination, the Unconventional Development in the UAE ensured that this water, particularly from wells with high H_2S and elevated temperatures, was managed through a well-defined strategy. This strategy included 5 primary pillars:

- Ensuring safety and regulatory compliance during disposal
- Choosing suitable formations for water management
- Selecting and designing disposal wells
- Guaranteeing effective water treatment
- Developing an ecosystem in hydraulic fracturing operations



Figure 1—Australian storage tank utilised during Zipper Frac operation

Ensuring safety and regulatory compliance during disposal. Managing the disposal of produced water is a significant issue for unconventional operations in the UAE because of the large amounts of water containing various pollutants. These pollutants include hydrocarbons, chemicals, bacteria, and solids from complex stimulation techniques and difficult reservoir conditions. Therefore, the disposal process must adhere to strict safety and regulatory standards. The main steps in this process are as follows:

- Treating the water to remove or reduce harmful substances to meet injection requirements. This step may involve filtration, chemical treatment, biological treatment, or thermal treatment, depending on the water's quality.
- Transporting the treated water to disposal wells using pipelines, trucks, or tanks. This phase needs careful monitoring, maintenance, and inspection of the equipment and infrastructure to prevent leaks, spills, or accidents, with transportation costs and logistics differing based on the distance between production wells and disposal wells.
- Injecting the water into disposal wells at controlled rates and pressures to avoid fracturing the formation or damaging the wellbore. This also involves regular testing and monitoring of the wells to ensure their integrity and performance, with injection rates and pressures determined by the formation's injectivity and the availability of disposal wells.
- Recording and reporting the volume, quality, and location of the disposed water to the relevant authorities and stakeholders. This step ensures transparency, accountability, and compliance with the regulations and permits governing water disposal.
- Evaluating and optimizing the disposal process to identify and address any issues, risks, or opportunities for improvement. This step involves analyzing the data collected from the previous steps, conducting audits and inspections, and implementing best practices and recommendations.

Formation Selection for Water Injection. Choosing the right formation for injecting treated water from unconventional operations is crucial. It needs sufficient porosity, permeability, and pressure to accept the water without affecting well integrity or reservoir quality. Additionally, it should be isolated from freshwater sources and hydrocarbon zones to avoid contamination. The Simsima formation, a long-used water disposal reservoir in conventional fields, has been selected as a suitable formation. This thick carbonate formation lies below oil and gas reservoirs and above basement rocks. **Fig. 2** illustrates the general stratigraphy and the position of the Simsima formation.

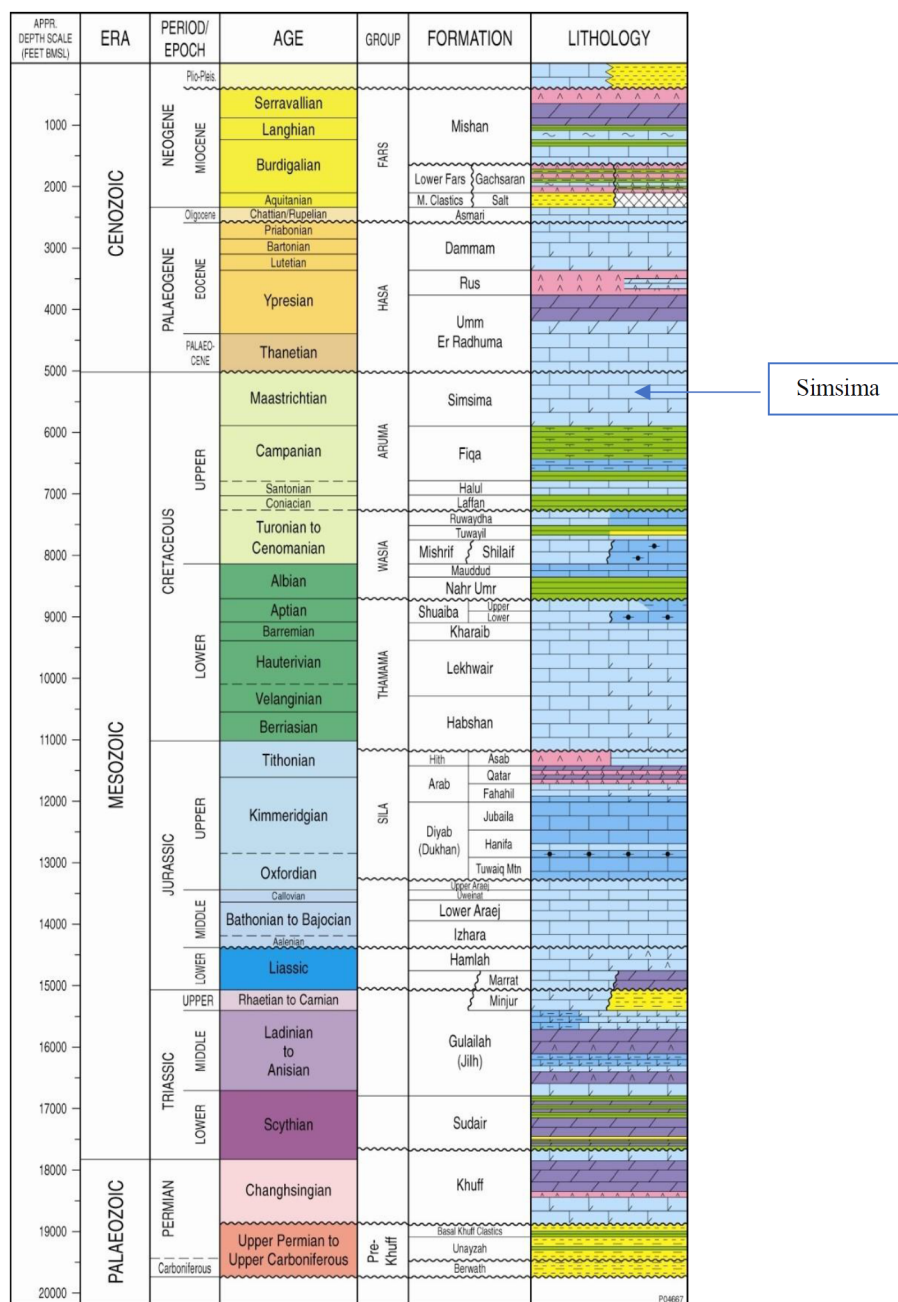


Figure 2—U.A.E general stratigraphy- Simsima Formation

Disposal Well Selection. Disposal wells serve the purpose of injecting treated water deep underground, ensuring it remains isolated from freshwater sources and hydrocarbon zones. These wells play a vital role in managing excess water in unconventional operations. Unconventional development thrives under stringent standards for safely and effectively designing such wells, choosing the Simsima formation as a dependable water disposal reservoir. The Simsima formation consists of a thick carbonate layer located below oil and gas reservoirs and above basement rocks, providing appropriate porosity, permeability, and pressure to accommodate the water without compromising well integrity or reservoir quality. **Fig.2** illustrates the general stratigraphy of the UAE and the location of the Simsima formation.

Well#01, which was temporarily suspended, was later converted into a disposal well. A comprehensive integrity study was conducted on Well#01, using various tools and tests to examine the cement quality, casing condition, and injection zones. The results confirmed that Well#01 had a solid cement bond, intact

casing, and isolated injection zones. It also provided the maximum injection rate and pressure required to prevent formation damage. Fig. 3 shows the status of Well#01 after its conversion.

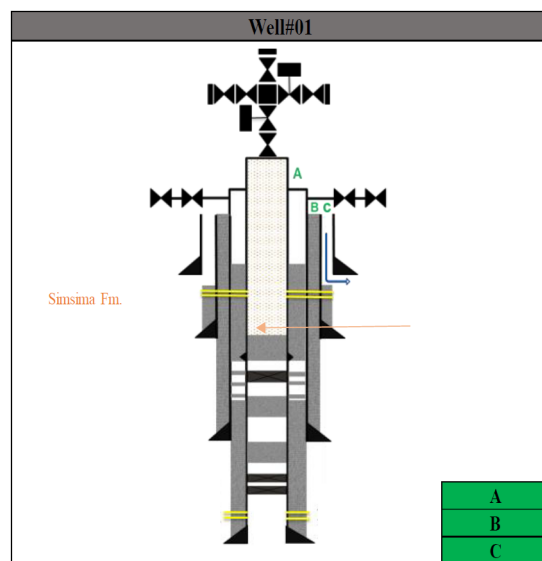


Figure 3—Status of Well#01.

Well integrity. Well integrity was evaluated using the Cement Bond Log (CBL) and tested to confirm the continuity of the cement and ensure isolation zones were in place to prevent any fluid migration to the surface and interaction with aquifers in the shallow zones behind the casing. The CBL also provided information on the cement quality and the casing condition, which are important factors for the long-term performance of the disposal well. The perforation and injectivity test were conducted to determine the optimal injection rate and pressure for the disposal well, as well as to identify any potential issues such as formation damage, plugging, or leakage. The injection test results were used to design the injection schedule and monitor the injection performance.

One of the primary methods to ensure well integrity in the disposal wells is by conducting Multi-Imaging Tool (MIT) regularly. MIT is a corrosion log that measures the changes in the casing diameter, which can indicate any corrosion or scale accumulation in the casing or the perforated part. These issues may require coil tubing clean out or acid injection to enhance the well injectivity. This log is run before the initial injection, following any workover, and at least annually. Performing MIT helps prevent potential risks such as fluid migration, casing failure, or environmental contamination. Fig. 4 shows an example of an MIT tool used for this purpose.

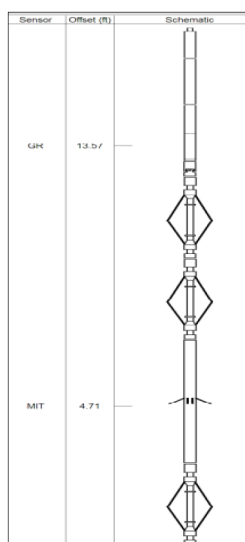


Figure 4—MIT Tool

Well#01 illustrates the potential for cost savings by transforming temporarily suspended wells into disposal wells. Additionally, a specialised design was created for the disposal well in the unconventional field, addressing the unique characteristics and challenges of the Simsima formation.

Well Injectivity tests. The well injectivity test was conducted during both the preparation and injection stages by monitoring the daily water injection into the formation and tracking the well's performance.

A secondary well was planned to stand by for the disposal Well#01 to manage the huge water volume and well integrity during high-volume water injections through the annulus and the perforated zone in the Simsima formation. Refer to [Tables 1, 2, and 3](#) below.

This system offers flexibility in opening and closing operations and supports continuous injection into disposal wells while maintaining well integrity.

Table 1—Treated water injection into Well#01 via the perforated zone within the Simsima formation.

Injection Well	Stat.	Inj. (Hrs)	WHFP (Psi)	A -ann (Psi)	C -ann (Psi)	Water rate (BWPD)
Well#1	O	24	17	0		1600
An Well#1	C	0			2	
An Well#2	C	0			23	

Table 2—Injection of treated water through Annulus C in Well#1.

Injection Well	Stat.	Inj. (Hrs)	WHP (Psi)	A -ann (Psi)	C -ann (Psi)	Water inj. (Bbl)
Well#1	C	0	120	0		
An Well#1	O	24			411.6	1633
An Well#02	C	24			-	

Table 3—Injection of treated water through annulus C in well #01 and well #02.

Injection Well	Stat.	Inj. (Hrs)	WHP (Psi)	A -ann (Psi)	C -ann (Psi)	Water inj. (Bbl)
Well#1	C	0	120	0		
An Well#01	O	24			400	1327
An Well#02	O	10			400	553

Water Injection challenges. Challenges commonly encountered during injection include the accumulation of scale due to high hardness and the precipitation of solids that obstruct the perforated sections of the well. These issues can reduce injectivity and compromise well integrity. To mitigate these problems, it is essential to closely monitor well performance and ensure they have clear access to handle large volumes of treated water. Regular maintenance, including coil tubing cleanouts and acid injections, can remove scale deposits and restore the wellbore. Additionally, it is crucial to consider the compatibility of injected water with formation water and rock, as well as potential risks like corrosion, souring, and microbiological growth. These factors can impact both injection efficiency and reservoir quality. Therefore, proper water treatment, continuous monitoring, and rigorous testing are vital for the success of the closed-loop system.

Guaranteeing effective water treatment.

Closed loop system

The treatment system was introduced to treat water from hydraulic fracturing in unconventional wells. It used a scavenger to lower hydrogen sulfide (H_2S), levels and a scale inhibitor to prevent scaling. The system included pipelines, tanks, pumps, and filters for water transport, storage, treatment, and injection. Sensors monitored injection parameters and detected any reservoir changes. Disposal wells were regularly inspected for safety and functionality. Injection rates and pressures were closely regulated to ensure safe and efficient operations. **Fig. 5** shows a simple schematic of the closed-loop system and the H_2S scrubber used before injection.



Figure 5—Closed loop-Sweetening system

Treated water Quality and Challenges. The flowback water received during the well test stage was very challenging to treat because of its high temperature and hydrogen sulfide (H_2S), concentration, which posed significant difficulties for the water quality and the treatment processes. To deal with this issue, a scavenger was added to the system to reduce the hydrogen sulfide (H_2S), level to below 100 ppm, which is the safety and regulatory limit for water injection into formations.

Water sampling and lab tests were performed regularly to confirm the accuracy and effectiveness of H_2S reduction in the treated water before disposal. One specific test involved using various types of scavengers to assess their efficiency in removing hydrogen sulfide (H_2S), from water that had high hardness, which was a common characteristic of produced water.

During the production phase, a highly effective chemical pumping system was created to optimise chemical injection for treating high H₂S water. Table 4 details the daily volumes of nitrogen (N₂) and H₂S Scavenger pumped, along with the water produced and its H₂S content, showing a reduction of H₂S to below 100 ppm during the field test.

Table 4—Field test for N₂ sparging and H₂S injection.

<i>N₂ sparging + H₂S scavenger Injection field test (September 2021)</i>					
<i>Days#</i>	<i>N₂ cons (l)</i>	<i>H₂S scav. (l/d)</i>	<i>WD (bbls/d)</i>	<i>Pur H₂S scav. per WD (l/bbl)</i>	<i>H₂S in water (ppm)</i>
1	2577	75	1053	0,071	40
2	2448	70	1169	0,060	40
3	3350	175	1382	0,127	10
4	2061	140	1409	0,099	60
5	2319	85	1000	0,085	40
6	4884	60	857	0,070	60
7	2448	60	1255	0,048	60
8	2061	55	1149	0,048	60
9	644	90	1264	0,071	60
10	1417	81	1265	0,064	60
<i>Average</i>	2421	89	1180	0,074	49
<i>Decreased N₂ injection to minimal (0.07MMSCFD) + H₂S scavenger injection rate at 10l/h</i>					
11	1160	99	1218	0,081	50
12	644	105	1292	0,081	70
<i>Average</i>	902	102	1255	0,081	60
<i>H₂S scavenger injection flowrate at KOD to 15-17 l/h and stopped N₂ injection</i>					
13	0	210	1294	0,162	70
14	0	140	1294	0,108	70
15	0	105	1142	0,092	70
16	0	110	1134	0,097	70
<i>Average</i>	0	141	1216	0,115	70
<i>Decreased N₂ injection to minimal (0.07MMSCFD) + H₂S scavenger injection rate at 15-17l/h</i>					
17	1288	115	1089	0,106	30
18	387	105	1097	0,096	30
<i>Average</i>	838	110	1093	0,074	30

Adjusting the concentration in relation to the water volume (bbls/d) and the injection rate reduced the hydrogen sulfide (H₂S), level to 30 ppm in the treated water. This outcome was accomplished through a mix of nitrogen (N₂) sparging and hydrogen sulfide (H₂S), scavenger, which exhibited a higher reaction rate and lower residual concentration compared to other scavengers.

The results of the field test demonstrated the feasibility and effectiveness of combining nitrogen (N₂) sparging and a hydrogen sulfide (H₂S), scavenger to treat high-hardness water for hydraulic fracturing. The treated water met the safety and regulatory standards for hydrogen sulfide (H₂S), content and also reduced the corrosion and scaling risks associated with high-hardness water. The closed-loop system also minimized the environmental impact and disposal costs of the produced water, making it a sustainable and economical solution for unconventional wells.

Beyond H₂S mitigation, the optimization of chemical dosing was further supported by real-time sensor feedback, enabling operators to swiftly respond to fluctuations in water quality and injection parameters.

This adaptive management approach ensured process efficiency and operational safety, reinforcing the reliability of the closed-loop system.

Furthermore, insights gained from laboratory studies and field trials encouraged the refinement of scavenger formulations to better suit the unique mineralogy and chemistry of UAE reservoirs. Collaboration between field engineers, chemists, and data analysts proved essential to achieving consistent water quality, especially under variable operational conditions.

In addition to addressing immediate technical challenges, the success of these initiatives has broader implications for the water management strategies in hydraulic fracturing. By integrating advanced treatment technologies and automated monitoring, operators can foster resource conservation, regulatory compliance, and environmental stewardship. Such improvements not only enhance well performance and longevity but also support the long-term sustainability goals of the energy sector in arid regions.

Looking forward, continued research into alternative scavenging agents, scaling inhibitors, and water reuse strategies is expected to further drive innovation, reduce costs, and lower the environmental footprint of unconventional development projects.

Developing an ecosystem in hydraulic fracturing operations

To achieve a zero-discharge target by reusing frac water, it is crucial to mitigate its high hardness and hydrogen sulfide (H_2S), contamination, ensuring treatment lowers hydrogen sulfide (H_2S), levels to 0ppm, meeting safety and regulatory standards for water injection. This requires a multi-stage process comprising filtration, scavenging, and scale inhibition. Utilizing treated water minimizes freshwater usage, decreases wastewater disposal impacts, and enhances both operational efficiency and sustainability in unconventional development. **Fig. 6** illustrates the ecosystem and zero discharge well concept.

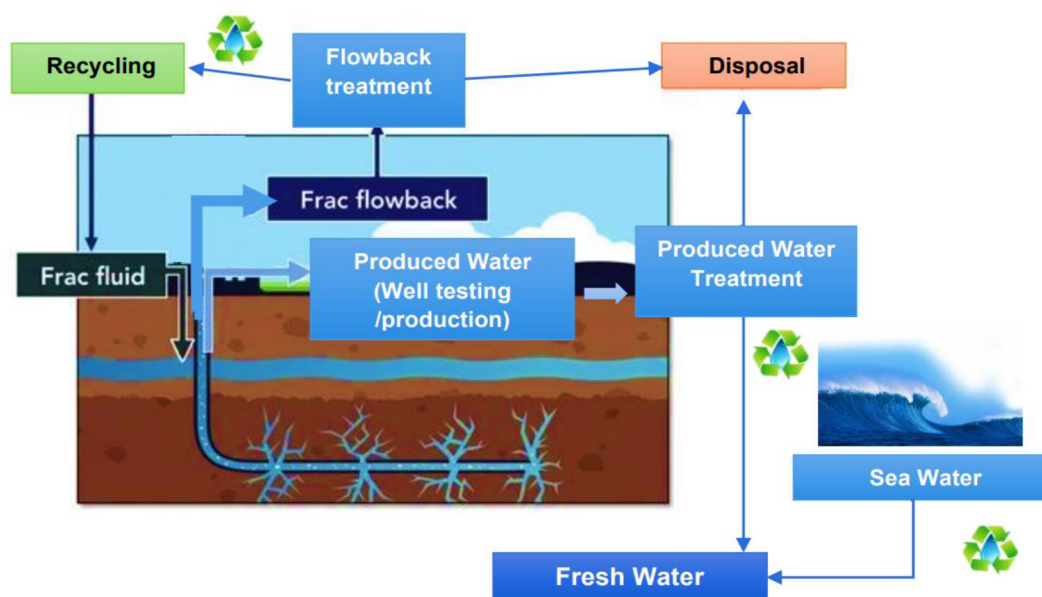


Figure 6—Ecosystem and Zero Discharge Well

The ecosystem and zero discharge well concept are a novel approach to achieve a closed-loop water cycle in hydraulic fracturing operations, where no water is discharged to the environment.

Challenges, Lessons Learned, and Future Opportunities

Some of the challenges, lessons learned, and future opportunities in the water management strategy for unconventional development in the UAE highlighted in this document are:

- The UAE experiences a high risk of water scarcity due to its arid climate, rapid population growth, and rising water demand. This situation calls for a comprehensive and integrated water management approach that addresses both quantity and quality.
- Unconventional development in the UAE requires substantial amounts of water for hydraulic fracturing, posing environmental and operational challenges. The UAE has introduced a closed-loop system to minimize freshwater consumption and wastewater production by maximizing the reuse of frac water and produced water.
- Reusing frac water and produced water necessitates innovative treatment technologies capable of managing high salinity, hardness, and hydrogen sulfide (H₂S), contamination. The UAE employs a multi-stage process involving filtration, scavenging, and scale inhibition to ensure the treated water complies with safety and regulatory standards for water injection.
- The treatment process also produces solid waste and residual brine that must be properly disposed of or repurposed. The UAE has investigated using these by-products as raw materials for industries like cement, fertilizer, or salt production, adhering to a circular economy model.
- The water management strategy for unconventional development in the UAE has yielded notable benefits, such as a 60% reduction in freshwater use, a 40% decrease in wastewater disposal, and improved operational efficiency and sustainability. Nonetheless, there remains potential for further optimization, including reducing the cost and energy consumption of the treatment process, increasing water and valuable mineral recovery rates, and broadening the scope and application of circular economy principles.

Conclusion

Demonstrating a pioneering spirit, the UAE has set a benchmark for sustainable management of unconventional resources by weaving together environmental stewardship, advanced technology, and economic sensibility. Through the strategic adoption of closed-loop water systems, the nation has not only minimized freshwater consumption but has also ensured that water used in hydraulic fracturing is treated and reused, substantially reducing environmental impacts.

Central to this achievement is the integration of state-of-the-art water treatment technologies designed to meet the unique challenges posed by high salinity, hardness, and contaminants such as hydrogen sulfide (H₂S). By implementing multi-stage processes—including sophisticated filtration, scavenging of hazardous substances, and scale inhibition—the treated water consistently meets stringent regulatory and safety standards for reinjection, supporting both operational efficiency and resource conservation.

Beyond water recycling, the UAE's vision extends to the responsible management of by-products. Instead of viewing residual brine and solid waste as disposal issues, these materials are now considered valuable resources. Initiatives are underway to repurpose these outputs in the production of cement, fertilizers, and industrial salts, echoing the principles of a circular economy and reducing the environmental footprint of oil and gas operations.

The results are compelling: a dramatic 60% reduction in freshwater use, a 40% decline in wastewater disposal, and tangible improvements in sustainability and operational effectiveness. Still, the journey continues. Opportunities abound for further innovation—whether in reducing the energy and cost demands of treatment processes, enhancing the recovery of water and minerals, or expanding the application of circular economy practices across the sector.

Altogether, the UAE's approach not only addresses immediate water scarcity challenges but also sets the stage for a resilient, resource-efficient future. This strategy aligns closely with the country's broader vision for water security and sustainable development, ensuring that the lessons learned here will inspire further advancements in resource management both locally and across the global energy industry.

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As we look to the future, we anticipate that the insights and best practices gained throughout this journey will guide our continued efforts in enhancing water management and resource development. The lessons learned are not merely milestones—they are stepping stones toward greater efficiency, environmental stewardship, and innovation. By fostering a culture of shared responsibility and continuous improvement, we are poised to further reduce our environmental footprint and create lasting value for both the industry and the broader community.

With each phase, we reaffirm our commitment to pioneering solutions that balance operational excellence with responsible resource utilization, ensuring that our approach remains aligned with the UAE's vision for sustainable development and water security.

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